

Working with Line Plots

Answer Key

Grade 3 — Data and Measurement

Quick Answers

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|--------------------------------------|--|
| 1. 4 | 6. 2 |
| 2. 12 | 7. 5 books |
| 3. 8 cm | 8. 32 books |
| 4. 4 cm | 9. 3 |
| 5. (b) how many marks in all = 10, — | 10. (a) the new most common length = 7 cm, — |
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What is verified

8 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 5, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3 — Data and Measurement
3.MD – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

4. *If they got 16: added the longest and shortest instead of subtracting*
6. *If they got 3: read off the 2-book column only and forgot to subtract the 1-book column*
8. *If they got 10: counted the marks instead of adding the values they sit above*
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Line Plot A (crayon lengths, cm): 6 has 2 marks, 7 has 3, 8 has 4, 9 has 2, 10 has 1 — twelve crayons in all.

Data Set B (books read): 3, 2, 4, 1, 2, 5, 3, 2, 6, 4 — ten students.

1. Use Line Plot A. How many crayons are 8 centimetres long?

Solution: Find 8 on the scale and count the marks stacked above it. There are four marks, and each mark is one crayon.

4

2. Use Line Plot A. How many crayons were measured altogether?

Solution: Every crayon put exactly one mark on the plot, so count all the marks: $2 + 3 + 4 + 2 + 1 = 12$. Counting the tick marks on the scale instead would give five, which is the number of different lengths, not the number of crayons. 12

3. Use Line Plot A. Which crayon length appears most often?

Solution: The most common value is the one with the tallest column of marks. The columns are 2, 3, 4, 2, 1 marks tall, and the tallest is the column of four sitting above 8.

8 cm

4. Use Line Plot A. How much longer is the longest crayon than the shortest one?

Solution: The longest crayon is the rightmost mark on the scale, 10 cm; the shortest is the leftmost, 6 cm. Subtract: $10 - 6 = 4$.

4 cm

5. Use Data Set B. (a) Build the line plot. (b) How many marks are on your finished plot?

Solution (a) — judged by eye. The scale runs 1 to 6, evenly spaced, and the marks stack in columns: one mark above 1, three above 2, two above 3, two above 4, one above 5, one above 6. Working through the list in order and crossing off each number as it is plotted is the reliable method; the tell-tale error is marks bunched sideways instead of stacked, which hides which column is tallest.

Solution (b): One mark per student, and Data Set B lists ten students, so the finished plot carries ten marks. This is also the check that the plot is complete. 10

6. Use the plot of Data Set B. How many more students read 2 books than read 1 book?

Solution: Compare two columns rather than reading one. The column above 2 has three marks; the column above 1 has one mark. “How many more” means subtract: $3 - 1 = 2$.

2

7. Use Data Set B. What is the difference between the greatest and the least number of books read?

Solution: The greatest value with any marks above it is 6; the least is 1. The difference is $6 - 1 = 5$. On the plot this is the distance from the leftmost occupied column to the rightmost.

5 books

8. Use Data Set B. How many books did the ten students read altogether?

Solution: This time the marks are not the answer — each mark stands for a student, and each student read the number of books its column sits above. Add the ten values: $3 + 2 + 4 + 1 + 2 + 5 + 3 + 2 + 6 + 4 = 32$. Column by column it is the same total: $1 + 6 + 6 + 8 + 5 + 6 = 32$.

32 books

9. Use Line Plot A. How many more crayons are 8 cm long than are 10 cm long?

Solution: The column above 8 has four marks and the column above 10 has one: $4 - 1 = 3$. 3

10. Two more crayons, each 7 cm long, are added to Line Plot A. (a) Which length is most common now? (b) Explain what changed.

Solution (a): The column above 7 had three marks; adding two more makes five. The column above 8 still has four. Five beats four, so the tallest column — and therefore the most common length — is now 7 cm. 7 cm

Solution (b) — instructor-judged. A full-credit answer says the column above 7 grew by two marks to five, which is now taller than the four marks above 8, so the most common length moved from 8 cm to 7 cm. Model answer: “The 7 column went from 3 marks to 5 marks. That is more than the 4 marks over 8, so the tallest column is now over 7. I only had to look at the two tallest columns, not recount them all.” An answer that names 7 cm without saying which column became the tallest, or without mentioning the two new marks, is not full credit.